From Songlines to the Cloud

Episode 3: Geography Gets a Chair

[Cold open]

Cambridge, 1672. Isaac Newton is about thirty years old. He's already the Lucasian Professor of Mathematics, and the work that will become the *Principia* is somewhere in his future. So what is he doing?

He's editing a geography textbook.

The book was the *Geographia Generalis*, by Bernhardus Varenius, a young German scholar working in Amsterdam who published it in 1650 and died the same year, at twenty-eight. It organized geography on mathematical and physical principles, and it split the subject in two: *general* geography, the laws that apply to the whole Earth, and *special* geography, the description of particular places. That split shaped the discipline for the next two centuries.

Newton didn't just put his name on it. He produced two Latin editions, in 1672 and 1681, redrew the original figures, and added some thirty new geometric diagrams of his own. Another Cambridge scholar, James Jurin, put out a further edition in 1712. Historians have used those editions to show that geography sat close to the center of the curriculum at some English universities around 1700.

[Beat]

So, before anyone held a chair in it, before there was a single department, geography was being taught at Cambridge from a textbook revised by the greatest scientist of the age. Remember that the next time someone tells you geography isn't a serious subject.

[Theme]

This is *From Songlines to the Cloud*. I'm Paul. At the end of the last episode, the Earth was finally measurable, navigable and mappable on a single grid. In this one, geography builds itself a home: university chairs, learned societies, journals, a place on the school timetable. It's also the episode in which the young discipline makes some choices, intellectual and moral, that it will spend the next century paying for.

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If Newton gave geography a textbook, Immanuel Kant gave it a philosophy.

Starting in the summer of 1756, Kant lectured on physical geography at the University of Königsberg. He taught the course forty-nine times over forty years, until July 1796. Only logic and metaphysics did he teach more often.

Kant's argument went like this. You can organize knowledge logically, by concept. Or you can organize it physically, and there are exactly two ways to do that: in time, which is history, or in space, which is geography. Geography, for Kant, was the empirical "knowledge of the world" that everything else rests on. A century and a half later, that idea would be picked up and leaned on very hard by the people who defined academic geography in the twentieth century.

Two cautions belong here. First, the popular claim that Kant was the first person to teach geography at a university is overstated; Cambridge was teaching from Varenius back in the 1670s. What Kant gave the subject was a sustained program and a philosophical defense. Second, and more important: Kant's geography lectures contain explicitly racist material, and recent scholarship treats them as central to debates about race and Eurocentrism at the very foundations of the discipline. If you teach the "time and space" idea, and it's a good idea, it's worth knowing what came packaged with it.

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The two people usually named as the founders of modern geography both died in the same year, 1859.

Alexander von Humboldt was the great traveler and the great generalist. His expedition to the Americas, from 1799 to

1804, made him famous; he founded the study of plant geography; and late in life he distilled his whole program into *Kosmos*, five volumes published between 1845 and 1862, the last after his death. It grew out of lectures he gave in Berlin. Humboldt wanted an empirical, quantitative, instrument-based science of the Earth as one interconnected system. The historian Susan Faye Cannon coined a phrase for it, "Humboldtian science," which she described as "astronomy and the physics of the earth and the biology of the earth all viewed from a geographical standpoint." Anyone who has ever stacked a climate layer on a vegetation layer on an elevation layer is, in a sense, doing Humboldtian science.

Carl Ritter was his counterpart: regional, comparative, and astonishingly persistent. In 1817 he published the first volume of his *Erdkunde*, a geography of the Earth in relation to nature and human history. It grew to twenty-one volumes and more than twenty thousand pages, and when he died it covered only Asia and Africa. Ritter was appointed at the University of Berlin in 1820 and held the first German chair of geography until his death. In 1828 he founded the Berlin Geographical Society, with Humboldt as a founding member. His framework was teleological, everything pointing toward a purpose, and organic. He saw the Earth as "a cosmic individual," its regions as organs, the whole thing expressing a divine plan.

Humboldt, systematic. Ritter, regional. Both gone in 1859, which conventionally marks the end of geography's classical period.

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What happened next was, above all, a project of the state.

In 1871 Oscar Peschel took up a new chair of geography at Leipzig, only the fourth in Germany. Then, in 1874, the Prussian government decided to establish a chair of geography at every Prussian university. That decree was the decisive moment in making geography an academic discipline. The motives were practical. Prussia had just won the Franco-Prussian War, the military value of geographic knowledge had been made very clear, there was a new nation to build, and the schools needed geography teachers.

There was one awkward detail. The people appointed to the new chairs were often trained in something else, the zoologist Friedrich Ratzel being the classic example, who then had to work out what exactly their new subject *was*. That question, what is geography, actually, will haunt this whole series from here on.

By the mid-1880s, when the Royal Geographical Society sent John Scott Keltie to survey geographical education in Europe, he counted twelve chairs of geography in Germany. France had founded chairs at Caen, Bordeaux and Lyon within six years of its defeat in 1871, and there were state-endowed chairs in Switzerland, the Netherlands and Italy. Britain had none.

In France, losing the war turned geography into a patriotic project. Paul Vidal de la Blache built the French school; he founded the journal *Annales de Géographie* in 1891 and took a chair at the Sorbonne in 1898. His idea of "possibilism," that the environment offers possibilities rather than dictating outcomes, would become the main counterweight to what was happening in Germany. More on that in a moment.

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Alongside the universities came the geographical societies, which mixed science, exploration, empire and a large paying public. Paris founded the first, in 1821. Berlin followed in 1828, London's Royal Geographical Society in 1830, the Russian Geographical Society in 1845, and the American Geographical Society in New York in 1851. By 1870 there were geographical societies in seven European countries plus Brazil, Mexico and the United States, and dozens more by 1890.

On the thirteenth of January 1888, thirty-three founders met at the Cosmos Club in Washington, D.C., and founded the National Geographic Society, "to increase and diffuse geographic knowledge." Gardiner Greene Hubbard was its first president. The first issue of the magazine came out that October.

And here's a detail that matters for the British story: it was the societies, not the universities, that paid to put geography back into English higher education. The Royal Geographical Society helped fund a readership at Oxford in 1887 and a lectureship at Cambridge in 1888.

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Now for the baggage.

The geographer Brian Hudson made an argument that has become a landmark. The "new geography" of the late nineteenth century, he said, was promoted largely to serve imperialism: territorial expansion, economic exploitation, militarism, and racist ideology. If you plot the growth of chairs, societies and school geography, it tracks the Scramble for Africa. David Livingstone's history of the discipline reaches a similar verdict: Victorian geography was deeply entangled with empire and with racial science.

Intellectually, the dominant idea of the era was environmental determinism: the belief that physical environment, especially climate, shapes human societies, and shapes them decisively.

Ratzel, the zoologist turned geographer, wrote a theory of the state as an organism and introduced the concept of *Lebensraum*, living space, in 1901. Decades later, by way of Karl Haushofer, that idea was taken up by Nazi geopolitics. His American student Ellen Churchill Semple opened her 1911 book *Influences of Geographic Environment* with a line that became the slogan of the whole school: "Man is a product of the earth's surface." And at Yale, Ellsworth Huntington made climate the determinant of "civilization" and of racial "character." He produced a map of "climatic energy" across the world, which peaked, conveniently, around his own New Haven.

[Beat]

It didn't last. Environmental determinism was attacked for its ethnocentrism, its racism, and its plain failure to fit the evidence. Franz Boas's anthropology, Vidal's possibilism, and Carl Sauer's cultural geography pushed it aside; historians date the turn to around 1920, and by mid-century the strong form was marginal.

But think about what that collapse cost geography as an institution. Determinism had been the discipline's big claim to discover laws, the thing that made it look like a science. When it fell, it took that claim with it, and it left an embarrassment that clung to the field for decades. Geography retreated into careful regional description. Hold on to that, because in the next episode it's going to matter a great deal.

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Meanwhile, the English-speaking world caught up.

The Keltie report, published in 1885 and 1886, documented how far Britain had fallen behind Germany. On the thirty-first of January 1887, a young Halford Mackinder stood up at the Royal Geographical Society and delivered "On the Scope and Methods of Geography," the manifesto of the "New Geography." He argued that geography could bridge "one of the greatest of all gaps," the gap "between the natural sciences and the study of humanity." That same year he became Reader in Geography at Oxford, the post partly funded by the Society, and that's effectively the start of continuous university geography in Britain. In 1904 he read a paper called "The Geographical Pivot of History," the Heartland thesis, and with it founded modern geopolitics. He also ran the London School of Economics and sat in Parliament. Mackinder is about as clear an example as you'll find of geography and imperial statecraft walking arm in arm.

In the United States, geography had been a staple of schooling since the early republic. Jedidiah Morse's *Geography Made Easy*, from 1784, was the first geography textbook published in the country, and it had sold at least four hundred thousand copies by the 1830s. Academic geography grew mostly out of geology. At Harvard, William Morris Davis developed his "cycle of erosion," which dominated American landform study for a generation, and in 1904 he led the founding of the Association of American Geographers. In 1903 the University of Chicago created the first separate geography department in the country to offer the PhD. For about twenty years it was the only full, independent geography department in America, and it trained the professors of the interwar years.

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And then there's school geography, which everywhere came before university geography, and which brings its own inheritance.

It entered the curriculum as "special geography": gazetteer facts about places, memorized. Capes and bays. The principal products of Peru. Rote learning of names, boundaries and exports, going back to the eighteenth century. In Britain, geography spread through elementary schools on a wave of imperial enthusiasm. Mackinder himself wrote school textbooks and campaigned for geography as training for imperial rule. Teachers organized in the Geographical Association, founded in 1893, and between the wars they gradually pushed capes and bays aside in favor of regional

study and fieldwork. In the United States, geography was a core subject of the nineteenth-century common school, usually taught by recitation.

[Beat]

If you teach geography today, you already know where this goes. That old rote image became the stereotype. Geography means memorizing capitals. And that stereotype, as we'll see, became one of the justifications for pushing the subject aside.

[Optional: a line of your own here, on the first time someone told you geography was "just memorizing capitals."]

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Let's close in 1939, with a book that defined American geography for the next fifteen years: Richard Hartshorne's *The Nature of Geography*. Drawing heavily on the German geographer Alfred Hettner, and through him on Kant, Hartshorne defined the field as "areal differentiation": the "accurate, orderly, and rational description and interpretation of the variable character of the earth surface." Geography, in this view, studies the unique character of regions. It describes and interprets. It does not go looking for general laws.

It's a thoughtful, careful position, and it shaped a generation. But put the pieces together. By 1939, American geography's one big scientific idea had spent two decades being discredited as racist pseudoscience. Its official answer to the question "what is geography?" was careful description of unique places. And in American schools, as we'll see, the subject had already been folded into something else.

Respectable. Serious. And, as it turned out, almost defenseless.

Next time: how America lost geography. It starts with a committee report in 1916, it comes to a head at Harvard in 1948, and it ends with young Americans who, on the eve of a war, couldn't find Iraq on a map.

I'm Paul. This has been *From Songlines to the Cloud*. Thanks for listening.

[Theme out]